FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – V
(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**D**	31.	**D**	61.	**C**
2.	**D**	32.	**D**	62.	**B**
3.	**C**	33.	**A**	63.	**C**
4.	**C**	34.	**D**	64.	**A**
5.	**C**	35.	**C**	65.	**D**
6.	**B**	36.	**A**	66.	**B**
7.	**A**	37.	**C**	67.	**A**
8.	**C**	38.	**D**	68.	**B**
9.	**D**	39.	**A**	69.	**D**
10.	**B**	40.	**A**	70.	**A**
11.	**A**	41.	**B**	71.	**C**
12.	**B**	42.	**C**	72.	**A**
13.	**D**	43.	**A**	73.	**C**
14.	**C**	44.	**B**	74.	**A**
15.	**C**	45.	**A**	75.	**A**
16.	**B**	46.	**D**	76.	**C**
17.	**B**	47.	**A**	77.	**B**
18.	**C**	48.	**A**	78.	**D**
19.	**A**	49.	**B**	79.	**B**
20.	**B**	50.	**D**	80.	**B**
21.	**D**	51.	**B**	81.	**B**
22.	**B**	52.	**D**	82.	**D**
23.	**A**	53.	**A**	83.	**B**
24.	**D**	54.	**D**	84.	**C**
25.	**C**	55.	**A**	85.	**D**
26.	**A**	56.	**A**	86.	**B**
27.	**B**	57.	**B**	87.	**B**
28.	**C**	58.	**A**	88.	**B**
29.	**D**	59.	**A**	89.	**B**
30.	**D**	60.	**A**	90.	**C**

Physics

PART – I

SECTION – A

1. **D**

$$P_t = P_c\left(1+\frac{m^2}{2}\right) = 9\left(1+\frac{(0.6)^2}{2}\right) = 9\left[1+\frac{0.36}{2}\right]$$

$$= 9[1+0.18] = 9[1.18] = 10.62 \text{ kw}$$

2. **D**

Out put of $G_1 = \overline{A}$, out put of $G_2 = \overline{B}$

Out put of G_3 = AB

Now $Y = \overline{AB}$(NAND gate)

3. **C**

$$N_1 = N_o e^{-10\lambda t}, N_2 = N_o e^{-\lambda t} \therefore \frac{N_1}{N_2} = \frac{1}{e^{9\lambda t}}$$

$$\therefore \frac{1}{e} = \frac{1}{e^{9\lambda t}} \therefore t = \frac{1}{9\lambda}$$

4. **C**

As the number of orbit increases the velocity decreases i.e K.E decreases the potential energy becomes less negative i.e P.E increase.

5. **C**

The current in the circuit decays monotonically.

6. **B**

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

Therefore, for a given mirror $\frac{1}{v}$ versus $\frac{1}{u}$ graph should be a straight line at u = f or

$\frac{1}{u}=\frac{1}{f}, v=\infty$ or $\frac{1}{v}=0$ and viceversa

7. **A**

Fact

8. **C**

$$\rho = \frac{\pi D^2}{4L}\frac{v}{i} = \frac{(3.14)\left(2.00\times10^{-3}\right)^{-2}}{4(0.314)}\left(\frac{100.0}{10.0}\right)$$

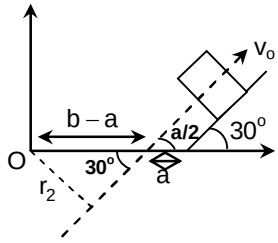
$$\rho = 1.00\times10^{-4}\Omega - m$$

9. **D**

$r_1 = (b-a)\sin 30^o$

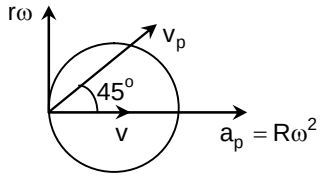
$= \left(\frac{b-a}{2}\right)$

$\therefore L = mv_o r_2 = \frac{mv_o(b-a)}{2}$



10. **B**

Angle is 45°



11. **A** $a = g\sin\theta - (\mu_o g\cos\theta)x \therefore \int_o^v vdv = \int_0^{x_{max}} \left[g\sin\theta - (\mu_o g\cos\theta)x\right]dx$

After solving we get $x_{max} = \frac{2\tan\theta}{\mu_o}$

12. **B**

$\frac{d\vec{v}}{dt} = ti + 3t^2 j \quad \therefore \int_0^{\vec{v}} d\vec{v} = \int_0^2 \left(ti + 3t^2 j\right)dt$

$\therefore \vec{v} = (2i + 8j)m/s.$

$\therefore v = \sqrt{68}$

by work energy theorem $\therefore w = \frac{1}{2}mv^2 \therefore w = 34J$

13. **D**

Net external force on the two blocks (whether they move with same retardation or not)

$f_{ext} = (0.2)(2+1)\times 10 = 6N$

$\therefore a_{cm} = \frac{f_{ext}}{2+1} = 2\ m/s^2$

14. **C**

Normal thrust per unit area is actually the pressure due to liquid

15. **C**

$PV = nRT \therefore pdv = nRdT \therefore dv = \left(\frac{v}{T}\right)dT \qquad .....(i)$

$dv = \gamma v dT \qquad .....(ii)$

from $(i)\ \&\ (ii)\ \gamma = \frac{1}{T}$

16. **B**

$v = \sqrt{gh} \therefore \frac{dh}{dt} = \sqrt{gh} \therefore t = \int_0^h \frac{dh}{\sqrt{gh}}$ or $t = 2\sqrt{\frac{h}{g}}$

Now at the time of meeting,
Time of fall of particle = time of wave pulse on reaching up to these

$\sqrt{\frac{2(L-h)}{g}} = 2\sqrt{\frac{h}{g}} \therefore h = \frac{L}{3}$

17. **B**

Comparing the given equation with $x = a\cos\omega t$

$\therefore \omega = \frac{\pi}{2}$ or $\frac{2\pi}{T} = \frac{\pi}{2} \therefore T = 4\,\text{sec}$

The given time t = 3 sec is really $\frac{3T}{4}$.

$\therefore$ distance covered will be 3 a

18. **C**

for $r \le r_1$ $V = \frac{Gm_1}{r_1} - \frac{Gm_2}{r_2} = \text{constant}$

for $r_1 \le r \le r_2$ $V = -\frac{Gm_2}{r_2} - \frac{Gm_1}{r_1}$

slope of V - x graph $dV/dr = Gm_1 / r^2$

for $r \ge r_2$ $V = -\frac{G}{r}(m_1 + m_2)$

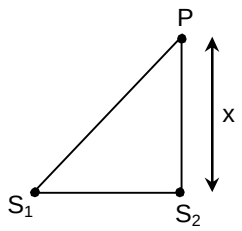
slope of V - r graph = $\frac{G}{r}(m_1 + m_2)$

At the boundary of outer shell slope of V – r graph changes from

$\frac{Gm_1}{r_2^2}$ to $\frac{G(m_1 + m_2)}{r_2^2}$ i.e slope increses

19. **A**

Path difference at S_2 is 2λ. Therefore for minimum intensity at P.



$S_1P - S_2P = \frac{3\lambda}{2}$ .....(i)

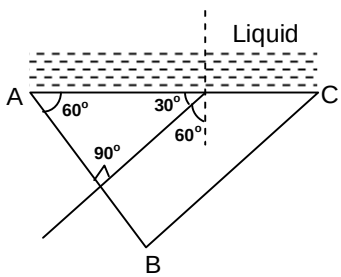
or $\sqrt{4\lambda^2 + x^2} - x = \frac{3\lambda}{2}$

Solving equation, $x = \frac{7\lambda}{12}$

20. **B**

Critical angle between glass and liquid face is $\sin\theta_C = \frac{\mu}{3/2}$

$\therefore \sin\theta_C = \frac{2\mu}{3}$



Angle of incidence at face AC is 60°
For TIR to take plane I > θ_C

Or $\sin 60^\circ > \frac{2\mu}{3}$ or $\mu < \frac{3\sqrt{3}}{4}$.

21. **D**

The two slabs will shift the image a distance

$$d = 2\left(1-\frac{1}{\mu}\right)t = 2\left(1-\frac{1}{1.5}\right)(1.5) = 1\,\text{cm}$$

22. **B**

The given network is a balanced Wheatstone bridge with one capacitor in parallel with this bridge.

23. **A**

No current will flow through the resistance at steady state.

24. **D**

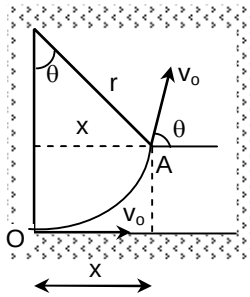
Series and parallel grouping of resistors.

25. **C**

$$r = \frac{mv_o}{B_o q} = \frac{v_o}{B_o \alpha}$$

$$\frac{x}{r} = \frac{\sqrt{3}}{2} = \sin\theta \therefore \theta = 60^o$$

$$t_{oA} = \frac{T}{6} = \frac{\pi}{3B_o\alpha}$$



Therefore x – coordinate of particle at any time $t > \frac{\pi}{3B_o\alpha}$ will be

$$x = \frac{\sqrt{3}}{2}\frac{v_o}{B_o\alpha} + v_o\left(t-\frac{\pi}{3B_o\alpha}\right)\cos 60^o \text{ So } x = \frac{\sqrt{3}}{2}\frac{v_o}{B_o\alpha} + \frac{v_o}{2}\left(t-\frac{\pi}{3B_o\alpha}\right)$$

26. **A**

$$B = 2\left[\frac{\mu_o}{4\pi}.\frac{i}{d/\sqrt{2}}\right]\left(\sin 90^o - \sin 45^o\right)$$

$$B = \frac{\mu_o i}{\sqrt{2}\pi d}\left(1-\frac{1}{\sqrt{2}}\right)\otimes$$

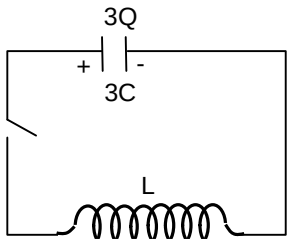
27. **B**

Polarity of emf will be opposite in the two cases while entering and while leaving the coil only in option (b) polarity is changing

28. **C**

The equivalent circuit is as shown in fig.

$$U_{L(max)} = U_{C(max)} = \frac{1}{2}\frac{(3Q)^2}{3C} = \frac{3Q^2}{2C}$$



29. **D**

A solid with completely filled valence band is an insulator, if the energy gap between the valence band and the empty conduction band is larger than about 5 eV

30. **D**

$A = 2\pi Rh = 2(\pi)\left(6.4\times10^{6}\right)(150)m^{2}$

$= 1.96\pi\times10^{3}km^{2}$

Chemistry

PART – II

SECTION – A

31. **D**

$$\text{Molarity} = \frac{d \times 1000}{Mw} = \frac{1.37 \times 1000}{68.5}$$

$$= \frac{125 \times 1000}{6250} = 20M$$

Note: In this question, % by wt should be given but it is missing therefore for calculation of molarity we consider 100% pure liquid.

32. **D**

$$HOCl \rightleftharpoons OH^- + Cl^+$$

C

$C(1-\alpha) \quad C\alpha \quad C\alpha$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.04 \times \alpha^2}{1-\alpha} \quad \begin{bmatrix} \text{As} & \alpha <<< 1 \\ \because & 1-\alpha \simeq 1 \end{bmatrix}$$

$$1.25 \times 10^{-4} = 0.04\alpha^2$$

$$\alpha^2 = \frac{1.25 \times 10^{-4}}{0.04} = \frac{125 \times 10^{-4}}{4}$$

$$\alpha = \frac{11}{2} \times 10^{-2} = \frac{0.11}{2} = 0.055$$

33. **A**

$$X \rightarrow 12 \times \frac{1}{6} + 3 + 2 \times \frac{1}{2} = 6$$

$$Y = 12 \times \frac{1}{4} = 3 \quad X_6Y_3 \Rightarrow X_2Y$$

34. **D**

A, B are explained by quantum mechanical model, C is explained by valence bond theory.

35. **C**

For a reaction $A \rightarrow P$

$$\text{In}\ \frac{k_2}{k_1} = \frac{E_A}{R}\left[\frac{1}{200} - \frac{1}{220}\right] = \ln 3$$

$$\text{In}\ \frac{k_2'}{k_1'} = \frac{E_B}{R}\left[\frac{1}{200} - \frac{1}{220}\right] = \ln 9$$

$$\frac{\ln 3}{\ln 3^2} = \frac{E_A}{E_B} \Rightarrow 2E_A = E_B$$

36. **A**

$$W_{isoreversible} = -2.303nRT\log\frac{V_2}{V_1}$$
$$= -2.303\times5\times8.314\times400\times\log2$$
$$= -2.303\times5\times8.314\times400\times0.3J$$
$$= -11488.28J$$

37. **C**

$$AgCl \rightleftharpoons \underset{s}{Ag^{+}}(aq.) + \underset{s}{Cl^{-}}(aq.)$$
$$K_{sp} = S^2 = 2.56\times10^{-10} = 256\times10^{-12}$$
$$S = 16\times10^{-6} = 1.6\times10^{-5}\,mol/L$$
$$1.6\times10^{-5} = \frac{0.01}{Vol(L)}$$
$$V = \frac{0.01}{1.6\times10^{-5}} = 625L$$

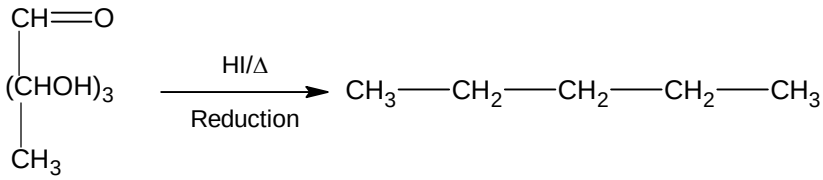
38. **D**

Aquaregia $\Rightarrow$ $HNO_3 + HCl[1:3]$

Pt + Aquaregia $\rightarrow$ $H_2[PtCl_6]$

Au + Aquaregia $\rightarrow$ $H[AuCl_4]$

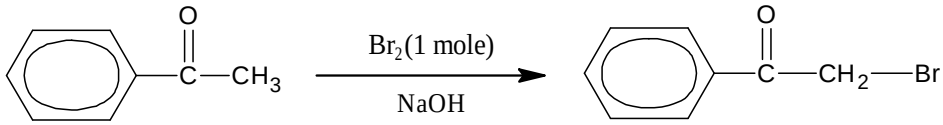
39. **A**



40. **A**

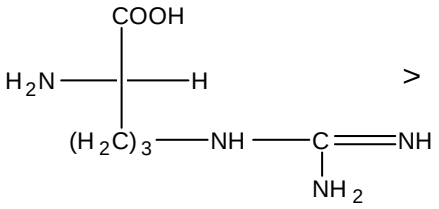
$$Cu_2S \longrightarrow Cu_2O$$

41. **B**

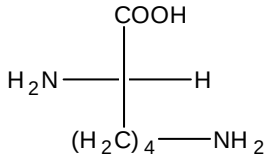


Halogenation reaction

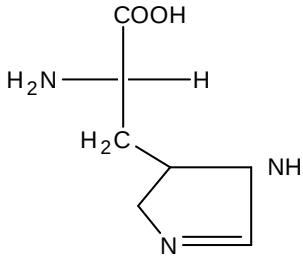
42. **C**



Arginine



Lysine

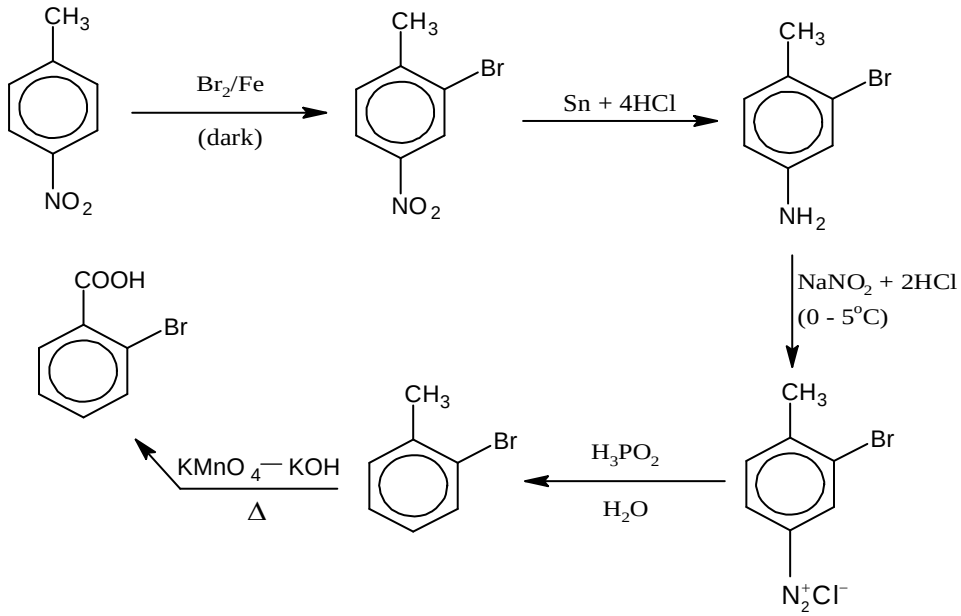


Histidine

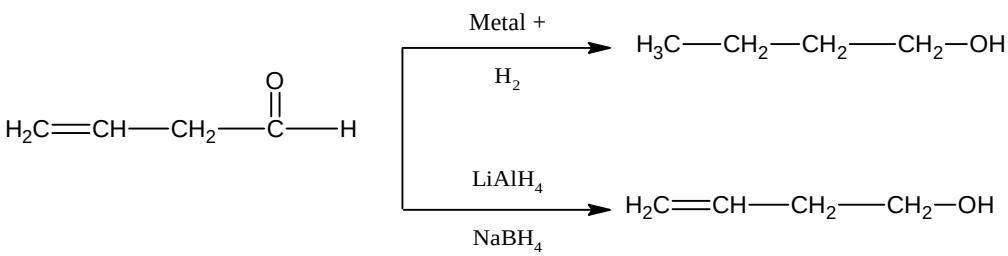
43. **A**

Nucleophilicity of S^- is more as it is weak base.

44. **B**



45. **A**



46. **D**

$[M(H_2O)_2(NH_3)_2Cl_2)]$ - complex show optical isomerism

47. **A**

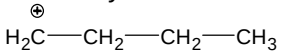
Not possible due to free rotation around C-C bond.

48. **A**

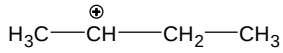
Electron withdrawing group increases acidic strength.

49. **B**

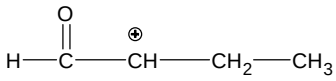
Reactivity towards SN^1 α stability of carbocation.



(I)



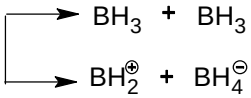
(II)



(III)

50. **D**

B_2H_6 + Lewis base



51. **B**

($FeSO_4$ is used)

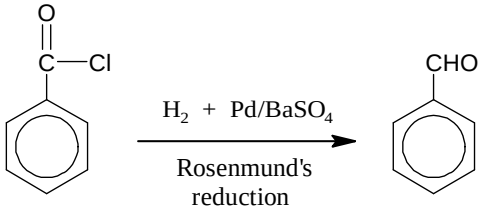
$Fe^{+2} + NO_3^- \longrightarrow F_e^{+3} + NO$

Brown ring complex $\left[Fe(H_2O)_5NO\right]^{+2}$

52. **D**

β-form has higher m.p. than α-form.

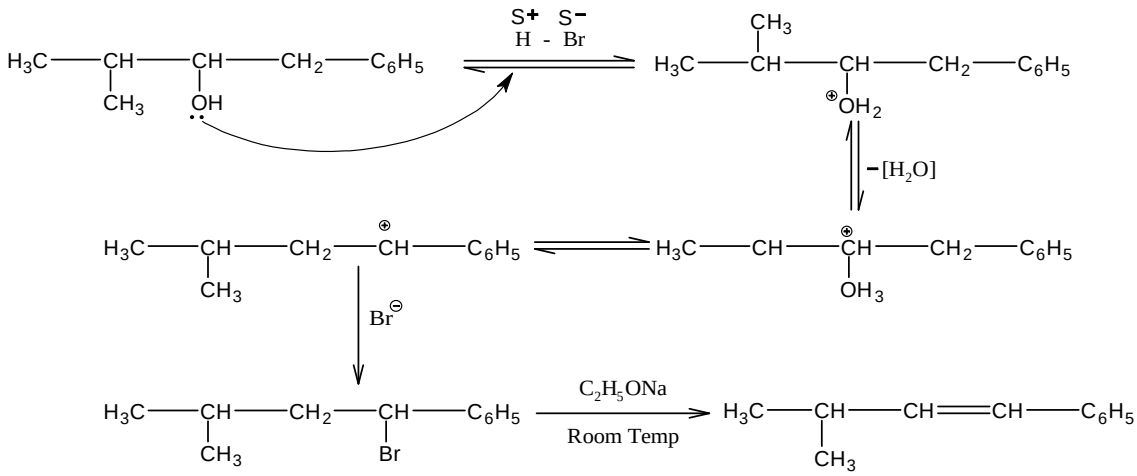
53. **A**



54. **D**

Fact.

55. **A**



56. **A**

$$\frac{r_{void}}{R} = 0.414$$

57. **B**

$$\log k = 14 - \frac{1.2\times10^4}{T} \text{.....(i)}$$

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$t_{1/2} = 6.93\times10^{-3}\ \text{min}$$

$$k = \frac{0.693}{6.93\times10^{-3}} = 100$$

$$\log 100 = 14 - \frac{1.2\times10^4}{T}$$

$$\Rightarrow T = 1000K$$

58. **A**

$$A_2X_3 \rightleftharpoons \underset{2s}{2A^{+3}} + \underset{3s}{3X^{-2}}$$

$$K_{sp} = (2s)^2(3s)^3$$

$K_{sp} = 108\ S^5$

$$S = \left(\frac{1.08\times10^{-23}}{108}\right)^{1/5} = 10^{-5}\ \text{mol/L}$$

59. **A**

Higher the value of z, difficult is to compress a gas.

60. **A**

$$XeF_6 \longrightarrow XeF_5^+ + F^-$$

Mathematics

PART – III

SECTION – A

61. **C**

$(x-1)^3+8=0 \Rightarrow x-1=-2,-2\omega,-2\omega^2$

$\Rightarrow \alpha=-1,\ \beta=1-2\omega,\ \gamma=1-2\omega^2$

$\Rightarrow \sum\frac{\alpha-1}{\beta-1}=\frac{3}{\omega}$

62. **B**

$x^2-x(y+4)+y^2-4y+16=0$

$D\geq 0 \Rightarrow (y-4)^2\leq 0 \Rightarrow y=4$

63. **C**

1, abc, $a^2b^2c^2$ also in A.P.

$\Rightarrow 2abc=1+a^2b^2c^2 \Rightarrow (abc-1)^2=0$

Now A.M. $\geq$ G.M., we get $27a+8b+125c\geq 90$

64. **A**

$f(x)$ is odd

So, $\int_{-2}^{2} f(x)\left[\frac{1}{x^2+1}\right]dx=0$

$\Rightarrow \int_{-2}^{3} f(x)\left[\frac{1}{x^2+1}\right]dx=\int_{2}^{3} f(x)\left[\frac{1}{x^2+1}\right]dx=0$

65. **D**

$\lim_{n\to\infty}\sum\frac{r^2}{n^3+r^3}=\int_0^1\frac{x^2}{1+x^3}dx=\frac{1}{3}\ln 2$

66. **B**

Number of words $=\frac{\lfloor 5}{\lfloor 2}-\frac{\lfloor 4}{\lfloor 2}=48$

67. **A**

$-y^2-y=1\Rightarrow y^2+y+1=0 \Rightarrow y=\omega$

So, $\omega^{2018}+\frac{1}{\omega^{2018}}+1=\omega^2+\frac{1}{\omega^2}+1=0$

68. **B**

$A+B+C=\pi$

By expanding we get $\Delta=0$

69. **D**

$\int_0^3\frac{x^2}{1+|x|^3}\left(\frac{1+3^x}{3^x+1}\right)dx \Rightarrow \frac{1}{3}\ln 28$

70. **A**

$f(x+100)=f(x)$

So, required limit $=\frac{1}{3}+\frac{1}{3^2}+.....\infty=\frac{1}{2}$

71. **C**

$P(B\cap C)=\frac{3}{4}-\frac{1}{3}-\frac{1}{3}=\frac{9-8}{12}=\frac{1}{12}$

72. **A**

Using diagram it is equal to A'.

73. **C**

$R=\{(1,3)(1,5)(2,3)(2,5)(3,5)(4,5)\}$

$R^{-1}=\{(3,1)(5,1)(3,2)(5,2)(5,3)(5,4)\}$

Now, $ROR^{-1}=\{(3,3)(3,5)(5,3)(5,5)\}$

74. **A**

Let $S:(\sim p\ v\sim q)\vee r$

$\Rightarrow S:(\sim F\vee\sim T)\vee F$

$\therefore S:(T\vee F)\vee F=T$

Hence, (A) is correct.

75. **A**

$M=\frac{\sum x}{n}=6$

Now, $\sum(x-M)^2=\sum x^2-2M\sum x+\sum M^2$

$\sigma=\frac{\sum(x-M)^2}{n}=\frac{640}{10}=64\Rightarrow\sigma=8$

76. **C**

$1+\cot\theta-\cot\frac{\theta}{2}=\frac{-\left(\cot\frac{\theta}{2}-1\right)^2}{2\cot\frac{\theta}{2}}\le 0$

$\Rightarrow\cot\frac{\theta}{2}=1\Rightarrow\theta=\frac{\pi}{2}$

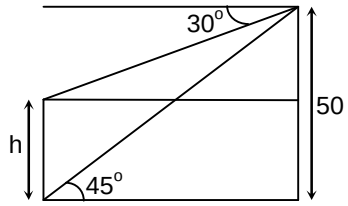
77. **B**

$a=\sqrt{b^2+c^2-2bc\cos A}=1$

$r=(s-a)\tan\frac{A}{2}=\frac{\sqrt{3}-1}{2}$

So, $\Delta=\pi r^2=\pi\left(1-\frac{\sqrt{3}}{2}\right)$

78. **D**



By using trigonometry, we get h = $50\left(1-\frac{\sqrt{3}}{3}\right)$ m

79. **B**

$A^2=\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}\Rightarrow$ A is nilpotent.

80. **B**

Let the required plane $by+cz+d=0$

Also, $-b+d=0\Rightarrow b=d$

81. **B**

Equation of tangent is $x+\sqrt{3}y=4$

So, $\Delta=\frac{1}{2}\times 1.\frac{4}{\sqrt{3}}=\frac{2}{\sqrt{3}}$ sq. unit

82. **D**

$f'(1)=0$ and $y(1)=1-\frac{\pi}{2}$

So, equation of tangent is $y=1-\frac{\pi}{2}$

83. **B**

It is symmetrical about x = 3.

84. **C**

Put $\tan\theta=t$

$\Rightarrow\int e^t\left(t^3+3t^2+t+1-1\right)dt=e^t\left(t^3+t\right)-e^t+c$

85. **D**

$x^4dx+\frac{x^3}{y^3}\left(\frac{ydx-xdy}{y^2}\right)=0\Rightarrow\frac{x^5}{5}+\frac{(x/y)^4}{4}=c$

86. **B**

$\left(y+\frac{2}{3}\right)^2=2\left(x-\frac{10}{9}\right)$

$Y^2=2X$

$X>2a,\ Y=0$

$x>\frac{19}{9},\ y=-\frac{2}{3}$

87. **B**

Let $\left(\sqrt{6}\cos\theta,\ \sqrt{3}\sin\theta\right)$ is point.

Then $\frac{x}{\sqrt{6}}\cos\theta+\frac{y}{\sqrt{3}}\sin\theta=1$ is parallel to x + y = 7

$\Rightarrow \frac{\cos\theta}{\sqrt{2}}=\frac{\sin\theta}{1}=\frac{1}{\sqrt{3}}$

Hence, $(a,\ b)\equiv(2,\ 1)$

88. **B**

$2\tan^{-1}\left(\frac{b}{a}\right)=\theta\Rightarrow\tan\frac{\theta}{2}=\frac{b}{a}$

So, $\cos\frac{\theta}{2}=\frac{a}{\sqrt{a^2+b^2}}=\frac{1}{e}$

89. **B**

$\left(\frac{z+\alpha\beta}{\alpha}\right)^3=1\Rightarrow z=\underset{A}{\alpha-\alpha\beta},\ \underset{B}{\alpha\omega-\alpha\beta},\ \underset{C}{\alpha\omega^2-\alpha\beta}$ are vertices.

Now, AB = BC = CA = $\sqrt{3}|\alpha|$

90. **C**

Number of subsets = ${}^{2n+1}C_1+{}^{2n+1}C_2+.....+{}^{2n+1}C_n$

$=\frac{2^{2n+1}}{2}-1=2^{2n}-1$